

IMPLEMENTATION

CODE:

```
CREATE DATABASE
```

```
CREATE DATABASE smart_water_utility;
```

```
USE smart_water_utility;
```

```
CREATE TABLE Customers (
```

```
    customer_id INT PRIMARY KEY AUTO_INCREMENT,
```

```
    customer_name VARCHAR(100) NOT NULL,
```

```
    phone VARCHAR(15),
```

```
    email VARCHAR(100),
```

```
    address VARCHAR(200)
```

```
);
```

```
CREATE TABLE Water_Consumption (
```

```
    consumption_id INT PRIMARY KEY AUTO_INCREMENT,
```

```
    customer_id INT,
```

```
    reading_date DATE,
```

```
    previous_reading INT,
```

```
    current_reading INT,
```

```
    units_consumed INT,
```

```
    FOREIGN KEY (customer_id)
```

```
        REFERENCES Customers(customer_id)
```

```
);
```

```
CREATE TABLE Bills (
```

```
    bill_id INT PRIMARY KEY AUTO_INCREMENT,
```

```
    customer_id INT,
```

```
    bill_date DATE,
```

```
    units_consumed INT,
```

```
    amount DECIMAL(10,2),
```

```
    due_date DATE,
```

```

    payment_status VARCHAR(20) DEFAULT 'Unpaid',

    FOREIGN KEY (customer_id)

        REFERENCES Customers(customer_id)

);

CREATE TABLE Complaints (

    complaint_id INT PRIMARY KEY AUTO_INCREMENT,

    customer_id INT,

    complaint_date DATE,

    complaint_type VARCHAR(100),

    description VARCHAR(300),

    status VARCHAR(30) DEFAULT 'Pending',

    FOREIGN KEY (customer_id)

        REFERENCES Customers(customer_id)

);

INSERT INTO Customers

(customer_name, phone, email, address)

VALUES

('Rahul Kumar', '9876543210', 'rahul@gmail.com', 'Chennai'),

('Priya Sharma', '9876543211', 'priya@gmail.com', 'Bangalore'),

('Arun Kumar', '9876543212', 'arun@gmail.com', 'Hyderabad'),

('Sneha Reddy', '9876543213', 'sneha@gmail.com', 'Chennai');

INSERT INTO Water_Consumption

(customer_id, reading_date, previous_reading,

current_reading, units_consumed)

VALUES

(1, '2026-08-01', 1000, 1025, 25),

(2, '2026-08-01', 2000, 2035, 35),

(3, '2026-08-01', 1500, 1520, 20),

(4, '2026-08-01', 3000, 3045, 45);

INSERT INTO Bills

```

```
(customer_id, bill_date, units_consumed,  
  
amount, due_date, payment_status)  
  
VALUES  
  
(1, '2026-08-01', 25, 250.00, '2026-08-15', 'Paid'),  
(2, '2026-08-01', 35, 350.00, '2026-08-15', 'Unpaid'),  
(3, '2026-08-01', 20, 200.00, '2026-08-15', 'Paid'),  
(4, '2026-08-01', 45, 450.00, '2026-08-15', 'Unpaid');
```

INSERT INTO Complaints

```
(customer_id, complaint_date, complaint_type,  
  
description, status)
```

VALUES

```
(1, '2026-08-05', 'Water Leakage',  
  
'Pipe leakage near house', 'Pending'),  
(2, '2026-08-06', 'Low Water Pressure',  
  
'Water pressure is very low', 'In Progress'),  
(3, '2026-08-07', 'Billing Issue',  
  
'Incorrect bill amount', 'Resolved'),  
(4, '2026-08-08', 'Water Supply',  
  
'No water supply', 'Pending');
```

SELECT * FROM Customers;

SELECT

```
c.customer_name,  
  
w.reading_date,  
  
w.previous_reading,  
  
w.current_reading,  
  
w.units_consumed
```

FROM Customers c

JOIN Water_Consumption w

ON c.customer_id = w.customer_id;

SELECT

```
c.customer_name,  
  
b.bill_id,  
  
b.units_consumed,  
  
b.amount,  
  
b.due_date,  
  
b.payment_status  
FROM Customers c  
  
JOIN Bills b  
  
ON c.customer_id = b.customer_id;  
  
SELECT  
  
c.customer_name,  
  
b.amount,  
  
b.due_date  
FROM Customers c  
  
JOIN Bills b  
  
ON c.customer_id = b.customer_id  
  
WHERE b.payment_status = 'Unpaid';  
  
SELECT *  
  
FROM Water_Consumption  
  
WHERE units_consumed > 30;  
  
SELECT  
  
c.customer_name,  
  
co.complaint_type,  
  
co.description,  
  
co.status  
FROM Customers c  
  
JOIN Complaints co  
  
ON c.customer_id = co.customer_id  
  
WHERE co.status = 'Pending';  
  
UPDATE Complaints
```

SET status = 'Resolved'

WHERE complaint_id = 1;

UPDATE Bills

SET payment_status = 'Paid'

WHERE bill_id = 2;

SELECT

SUM(amount) AS total_revenue

FROM Bills

WHERE payment_status = 'Paid';

SELECT

SUM(units_consumed) AS total_water_consumed

FROM Water_Consumption;

SELECT

c.customer_name,

SUM(w.units_consumed) AS total_consumption

FROM Customers c

JOIN Water_Consumption w

ON c.customer_id = w.customer_id

GROUP BY c.customer_id, c.customer_name;

OUTPUT:

```
MySQL 8.0 Command Line Cli  x  +  v
mysql> -- 1. CREATE DATABASE
mysql> CREATE DATABASE smart_water_utility;
Query OK, 1 row affected (0.01 sec)

mysql> USE smart_water_utility;
Database changed
mysql>
mysql> -- =====
mysql> -- 2. CUSTOMER TABLE
mysql> -- =====
mysql> CREATE TABLE Customers (
->  customer_id INT PRIMARY KEY AUTO_INCREMENT,
->  customer_name VARCHAR(100) NOT NULL,
->  phone VARCHAR(15),
->  email VARCHAR(100),
->  address VARCHAR(200)
-> );
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql>
mysql> -- =====
mysql> -- 3. WATER CONSUMPTION TABLE
mysql> -- =====
mysql> CREATE TABLE Water_Consumption (
->  consumption_id INT PRIMARY KEY AUTO_INCREMENT,
->  customer_id INT,
->  reading_date DATE,
->  previous_reading INT,
->  current_reading INT,
```

```

MySQL 8.0 Command Line Cli  x  +  v
mysql> -- =====
mysql> -- 4. BILLING TABLE
mysql> -- =====
mysql>
mysql> CREATE TABLE Bills (
->   bill_id INT PRIMARY KEY AUTO_INCREMENT,
->   customer_id INT,
->   bill_date DATE,
->   units_consumed INT,
->   amount DECIMAL(10,2),
->   due_date DATE,
->   payment_status VARCHAR(20) DEFAULT 'Unpaid',
->   FOREIGN KEY (customer_id)
->     REFERENCES Customers(customer_id)
-> );
Query OK, 0 rows affected (0.07 sec)

mysql>
mysql>
mysql> -- =====
mysql> -- 5. COMPLAINT TABLE
mysql> -- =====
mysql>
mysql> CREATE TABLE Complaints (
->   complaint_id INT PRIMARY KEY AUTO_INCREMENT,
->   customer_id INT,
->   complaint_date DATE,
->   complaint_type VARCHAR(100),
->   description VARCHAR(300),
->   status VARCHAR(30) DEFAULT 'Pending',
->   FOREIGN KEY (customer_id)
->     REFERENCES Customers(customer_id)
-> );

```

```

MySQL 8.0 Command Line Cli  x  +  v
mysql> -- =====
mysql> -- 6. INSERT CUSTOMER DATA
mysql> -- =====
mysql>
mysql> INSERT INTO Customers
-> (customer_name, phone, email, address)
-> VALUES
-> ('Rahul Kumar', '9876543210', 'rahul@gmail.com', 'C'),
-> ('Priya Sharma', '9876543211', 'priya@gmail.com', 'C'),
-> ('Arun Kumar', '9876543212', 'arun@gmail.com', 'Hyd'),
-> ('Sneha Reddy', '9876543213', 'sneha@gmail.com', 'C');
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql>
mysql>
mysql> -- =====
mysql> -- 7. INSERT WATER CONSUMPTION DATA
mysql> -- =====
mysql>
mysql> INSERT INTO Water_Consumption
-> (customer_id, reading_date, previous_reading,
->   current_reading, units_consumed)
-> VALUES
-> (1, '2026-08-01', 1000, 1025, 25),
-> (2, '2026-08-01', 2000, 2035, 35),
-> (3, '2026-08-01', 1500, 1520, 20),
-> (4, '2026-08-01', 3000, 3045, 45);
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql>
mysql>

```

```

MySQL 8.0 Command Line C#  X  +  v
mysql> -- =====
mysql> -- 8. INSERT BILL DATA
mysql> -- =====
mysql>
mysql> INSERT INTO Bills
-> (customer_id, bill_date, units_consumed,
-> amount, due_date, payment_status)
-> VALUES
-> (1, '2026-08-01', 25, 250.00, '2026-08-15', 'Paid'),
-> (2, '2026-08-01', 35, 350.00, '2026-08-15', 'Unpaid'),
-> (3, '2026-08-01', 20, 200.00, '2026-08-15', 'Paid'),
-> (4, '2026-08-01', 45, 450.00, '2026-08-15', 'Unpaid');
Query OK, 4 rows affected (0.01 sec)
Records: 4 Duplicates: 0 Warnings: 0

mysql>
mysql>
mysql> -- =====
mysql> -- 9. INSERT COMPLAINT DATA
mysql> -- =====
mysql>
mysql> INSERT INTO Complaints
-> (customer_id, complaint_date, complaint_type,
-> description, status)
-> VALUES
-> (1, '2026-08-05', 'Water Leakage',
-> 'Pipe leakage near house', 'Pending'),
->
-> (2, '2026-08-06', 'Low Water Pressure',
-> 'Water pressure is very low', 'In Progress'),
->
-> (3, '2026-08-07', 'Billing Issue',
-> 'Incorrect bill amount', 'Resolved'),

```

```

MySQL 8.0 Command Line C#  X  +  v
Query OK, 4 rows affected (0.00 sec)
Records: 4 Duplicates: 0 Warnings: 0

mysql>
mysql>
mysql> -- =====
mysql> -- 10. DISPLAY ALL CUSTOMERS
mysql> -- =====
mysql>
mysql> SELECT * FROM Customers;
+-----+-----+-----+-----+-----+
| customer_id | customer_name | phone | email | address |
+-----+-----+-----+-----+-----+
| 1 | Rahul Kumar | 9876543210 | rahul@gmail.com | Chennai |
| 2 | Priya Sharma | 9876543211 | priya@gmail.com | Bangalore |
| 3 | Arun Kumar | 9876543212 | arun@gmail.com | Hyderabad |
| 4 | Sneha Reddy | 9876543213 | sneha@gmail.com | Chennai |
+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql>
mysql>
mysql> -- =====
mysql> -- 11. DISPLAY WATER CONSUMPTION
mysql> -- =====
mysql>
mysql> SELECT
-> c.customer_name,
-> w.reading_date,
-> w.previous_reading,
-> w.current_reading,
-> w.units_consumed
-> FROM Customers c

```

```
MySQL 8.0 Command Line Cli x + v
-> w.reading_date,
-> w.previous_reading,
-> w.current_reading,
-> w.units_consumed
-> FROM Customers c
-> JOIN Water_Consumption w
-> ON c.customer_id = w.customer_id;

+-----+-----+-----+-----+-----+
| customer_name | reading_date | previous_reading | current_reading | units_consumed |
+-----+-----+-----+-----+-----+
| Rahul Kumar   | 2026-08-01   | 1000             | 1025            | 25             |
| Priya Sharma  | 2026-08-01   | 2000             | 2035            | 35             |
| Arun Kumar    | 2026-08-01   | 1500             | 1520            | 20             |
| Sneha Reddy   | 2026-08-01   | 3000             | 3045            | 45             |
+-----+-----+-----+-----+-----+

4 rows in set (0.00 sec)

mysql>
mysql>
mysql> -- =====
mysql> -- 12. DISPLAY CUSTOMER BILLS
mysql> -- =====
mysql>
mysql> SELECT
-> c.customer_name,
-> b.bill_id,
-> b.units_consumed,
-> b.amount,
-> b.due_date,
-> b.payment_status
-> FROM Customers c
-> JOIN Bills b
-> ON c.customer_id = b.customer_id;
```

```
MySQL 8.0 Command Line Cli x + v
-> ON c.customer_id = b.customer_id;

+-----+-----+-----+-----+-----+-----+
| customer_name | bill_id | units_consumed | amount | due_date | payment_status |
+-----+-----+-----+-----+-----+-----+
| Rahul Kumar   | 1       | 25             | 250.00 | 2026-08-15 | Paid           |
| Priya Sharma  | 2       | 35             | 350.00 | 2026-08-15 | Unpaid        |
| Arun Kumar    | 3       | 20             | 200.00 | 2026-08-15 | Paid           |
| Sneha Reddy   | 4       | 45             | 450.00 | 2026-08-15 | Unpaid        |
+-----+-----+-----+-----+-----+-----+

4 rows in set (0.00 sec)

mysql>
mysql>
mysql> -- =====
mysql> -- 13. DISPLAY UNPAID BILLS
mysql> -- =====
mysql>
mysql> SELECT
-> c.customer_name,
-> b.amount,
-> b.due_date
-> FROM Customers c
-> JOIN Bills b
-> ON c.customer_id = b.customer_id
-> WHERE b.payment_status = 'Unpaid';

+-----+-----+-----+
| customer_name | amount | due_date |
+-----+-----+-----+
| Priya Sharma  | 350.00 | 2026-08-15 |
| Sneha Reddy   | 450.00 | 2026-08-15 |
+-----+-----+-----+

2 rows in set (0.00 sec)
```



```
MySQL 8.0 Command Line Cli x + v
mysql> SELECT *
-> FROM Water_Consumption
-> WHERE units_consumed > 30;
+-----+-----+-----+-----+-----+-----+
| consumption_id | customer_id | reading_date | previous_reading | current_reading | units_consumed |
+-----+-----+-----+-----+-----+-----+
| 2 | 2 | 2026-08-01 | 2000 | 2035 | 35 |
| 4 | 4 | 2026-08-01 | 3000 | 3045 | 45 |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql>
mysql>
mysql> -- =====
mysql> -- 15. DISPLAY PENDING COMPLAINTS
mysql> -- =====
mysql>
mysql> SELECT
-> c.customer_name,
-> co.complaint_type,
-> co.description,
-> co.status
-> FROM Customers c
-> JOIN Complaints co
-> ON c.customer_id = co.customer_id
-> WHERE co.status = 'Pending';
+-----+-----+-----+-----+
| customer_name | complaint_type | description | status |
+-----+-----+-----+-----+
| Rahul Kumar | Water Leakage | Pipe leakage near house | Pending |
| Sneha Reddy | Water Supply | No water supply | Pending |
+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

```
MySQL 8.0 Command Line Cli x + v
mysql> UPDATE Complaints
-> SET status = 'Resolved'
-> WHERE complaint_id = 1;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql>
mysql>
mysql> -- =====
mysql> -- 17. UPDATE BILL PAYMENT STATUS
mysql> -- =====
mysql>
mysql> UPDATE Bills
-> SET payment_status = 'Paid'
-> WHERE bill_id = 2;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql>
mysql>
mysql> -- =====
mysql> -- 18. CALCULATE TOTAL REVENUE
mysql> -- =====
mysql>
mysql> SELECT
-> SUM(amount) AS total_revenue
-> FROM Bills
-> WHERE payment_status = 'Paid';
+-----+
| total_revenue |
+-----+
| 800.00 |
+-----+
```

```

MySQL 8.0 Command Line Cll  x  +  v
mysql> SELECT
->     SUM(units_consumed) AS total_water_consumed
-> FROM Water_Consumption;
+-----+
| total_water_consumed |
+-----+
|                125 |
+-----+
1 row in set (0.00 sec)

mysql>
mysql>
mysql> -- =====
mysql> -- 20. CUSTOMER-WISE TOTAL CONSUMPTION
mysql> -- =====
mysql>
mysql> SELECT
->     c.customer_name,
->     SUM(w.units_consumed) AS total_consumption
-> FROM Customers c
-> JOIN Water_Consumption w
-> ON c.customer_id = w.customer_id
-> GROUP BY c.customer_id, c.customer_name;
+-----+-----+
| customer_name | total_consumption |
+-----+-----+
| Rahul Kumar  |                25 |
| Priya Sharma |                35 |
| Arun Kumar   |                20 |
| Sneha Reddy  |                45 |
+-----+-----+
4 rows in set (0.00 sec)

```